Page Replacement Algorithms

CSE Sem-4

Page Replacement Policies

- For the concept of ‘multiprogramming’, if multiple processes has to be loaded in main memory at the same time, it has to be decided how many frames will be allocated to each process.
- When page replacement is required, is has to be selected which frame will be replaced.

- **First in first out (FIFO) :** Replaces the page which **has been in the memory for the longest period of time.**
- **Least recently used (LRU) :** Replaces the page which **has not been used for the longest period of time.**
- **Optimal page replacement (Opt) :** Replaces the page which **will not be used for the longest period of time.**

First In First Out (FIFO)

- The oldest page will be replaced during the time of page replacement.

  For example, 3 frames of main memory will be allocated for a process. Considering the frames are initially empty.

Reference String:

7	0	1	2	0*	3	0	4	2	3	0	3*	2*	1	2	0*	1*	7	0	1
7	7	7	2		2	2	4	4	4	0			0	0			7	7	7
	0	0	0		3	3	3	2	2	2			1	1			1	0	0
		1	1		1	0	0	0	3	3			3	2			2	2	1

Page Faults: 15

Least Recently Used (LRU)

- The page will be replaced that has not been used for the longest period of time.

  For example, 3 frames of main memory will be allocated for a process. Considering the frames are initially empty.

Reference String:

7	0	1	2	0*	3	0*	4	2	3	0	3*	2*	1	2*	0	1*	7	0*	1*
7	7	7	2		2		4	4	4	0			1		1		1		
	0	0	0		0		0	0	3	3			3		0		0		
		1	1		3		3	2	2	2			2		2		7		

Page Faults:12

Optimal Page Replacement (Opt)

- Replace the page that will not be used for the longest period of time.

  For example, 3 frames of main memory will be allocated for a process. Considering the frames are initially empty.

  Reference String:

7	0	1	2	0*	3	0*	4	2*	3*	0	3*	2*	1	2*	0*	1*	7	0*	1*
7	7	7	2		2		2			2			2				7		
	0	0	0		0		4			0			0				0		
		1	1		3		3			3			1				1		

  Page Faults : 9

- Optimal page replacement is difficult to implement, because it requires future knowledge of the reference string.